Unit 1

Lesson 2

Spark The Challenge!

Danger in the Air!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Describe how people with hearing or vision impairments may detect or miss environmental danger cues.
- Identify the specific barriers these individuals may face during emergency situations.
- Model basic chemical reactions that occur during gas leaks or combustion
- Demonstrate how heat transfers through conduction, convection, and radiation in fire-related scenarios.
- Convert measurements between units.
- Explain why scientific notation is important for expressing large or small values in emergency systems.

Challenge questions of the lesson.

- How do our senses help us notice signs of danger, and what might be harder to detect without sight or hearing?
- In what ways might a person with vision or hearing loss experience the world differently during an emergency?
- What barriers could someone with a disability face when trying to escape a dangerous situation, and how could these be reduced?
- How do emergency alert systems warn people about danger, and what makes them effective for everyone—including those with disabilities?
- What happens at the particle level during a gas leak or when something catches fire?
- How could we model a chemical reaction that might happen in a real-life emergency?
- When a fire spreads, how does heat move through objects, air, and radiation, and how could this affect escape routes?
- How can we read a chemical formula like CO_2 or CH_4 and understand what it means?
- Why is it important to convert between units quickly during an emergency?
- How can powers of ten help us compare very large and very small numbers in safety systems?
- Why do scientists and engineers use scientific notation when working with emergency sensors or warning systems?

SEL Relation

- **Self-Awareness:** Recognizing one's own biases and potential as a empathetic problem-solver.
- **Self-Management:** Practicing patience and organization when defining complex problems.
- **Social Awareness:** Developing empathy by understanding diverse user needs through personas, especially for assistive technology.
- **Relationship Skills:** Engaging in active listening and respectful communication, particularly when discussing user needs.
- **Responsible Decision-Making:** Considering ethical implications and user-centered approaches in design.

Engage

Teacher's Role:

- **Facilitator**
  - o Create a safe and inclusive space where all students feel comfortable sharing ideas.
  - o Use inclusive language and encourage multiple perspectives, especially from empathy-based viewpoints.
  - o Guide students to link the story to their own experiences or observations in real life.
- **Questioner**
  - o Ask open-ended questions that prompt deeper thinking, such as:

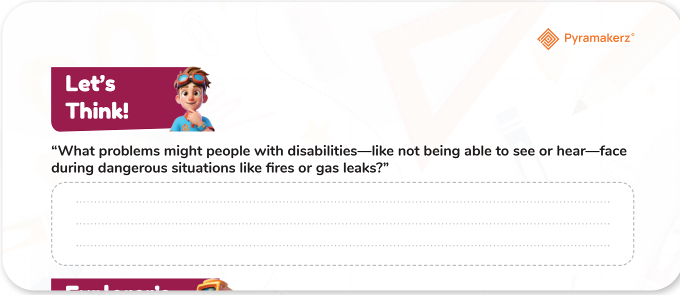
    "How would you feel if you couldn't see or hear during an emergency?"

    "What signals could still help you know something dangerous was happening?"
  - o Encourage "why" and "how" answers to promote reasoning.
- **Supporter**
  - o Offer encouragement to students who might be hesitant to share ideas.
  - o Rephrase or scaffold questions for learners who need additional support.
  - o Celebrate all contributions to build confidence and engagement.
- **Collaborator**
  - o Join in small group discussions to model respectful listening and idea-building.
  - o Help groups synthesize multiple viewpoints into shared insights.
- **Assessor**
  - o Listen carefully to students' verbal responses for understanding and empathy.
  - o Observe group participation, noting use of relevant vocabulary (e.g., senses, danger cues, accessibility).
  - o Provide feedback that reinforces accurate understanding and corrects misconceptions.

Expected From Students:

- Reflect on how the situation might feel for people with different abilities.
- Share observations, questions, and connections during class discussion.
- Use a thinking routine like See-Think-Wonder to structure ideas.
- Collaborate respectfully in pairs or groups to explore possible solutions.
- Record ideas in a graphic organizer, journal, or digital note-taking tool.
- Demonstrate empathy and problem-solving skills when suggesting ways to improve safety for all people.

Parts of the books included.


Pyramakerz®

Let's Think!

"What problems might people with disabilities—like not being able to see or hear—face during dangerous situations like fires or gas leaks?"

Let's Think!

Objective:

Students will identify challenges faced by people with vision or hearing impairments during emergencies such as fires or gas leaks, and propose strategies to make emergency alerts more inclusive and effective.

Answers (Sample student responses):

- People with vision loss may not see flashing lights or warning signs.
- People with hearing loss may not hear alarms, sirens, or verbal instructions.
- Narrow or blocked exits can make evacuation harder for people with mobility challenges.
- Crowded, noisy, or dark environments increase difficulty in detecting danger.
- Complex or unclear instructions can cause confusion in stressful moments.

Possible Strategies for Implementation:

- Install alarms that combine sound, light, and vibration to reach all senses.
- Use text-based emergency alerts on phones or public screens.
- Provide clear, high-contrast signage in emergency exits.
- Include tactile maps or guides in public spaces.
- Train emergency staff to communicate using multiple methods (spoken instructions, gestures, written notes).
- Run practice drills that include accessibility considerations for all participants.

Explorer's Toolkit

Teacher's Role:

- **Questioner**

  Use open-ended prompts to drive inquiry:

o "How could a machine help someone sense danger without relying on sight or hearing?"

o "What signals could we use instead of sound or light?"

Encourage students to predict and test their ideas.

- **Supporter**

  Model the thought process for designing a simple machine, provide sentence starters (e.g., “I think my machine will…”), and circulate to help students troubleshoot mechanical or safety issues.

- **Collaborator**

  Work alongside groups, test ideas together, and celebrate creative solutions. Foster peer-to-peer assistance and idea-sharing.

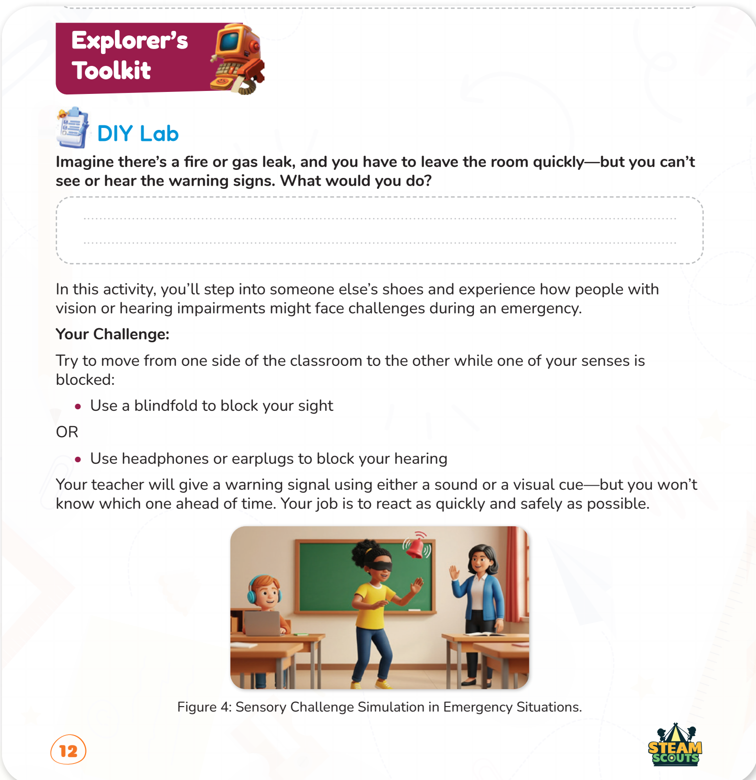
- **Assessor**

  Observe how students plan, build, and adapt their machines. Ask probing questions to assess understanding of the mechanics and empathy in design.

Expected From Students:

- Collaboratively design and build a simple mechanical prototype that can signal an emergency without relying on sight or hearing.
- Explain how their machine works and how it supports people with specific needs.
- Reflect on the trial-and-error process.

Parts of the books included.



Explorer’s Toolkit

DIY Lab

Imagine there’s a fire or gas leak, and you have to leave the room quickly—but you can’t see or hear the warning signs. What would you do?

In this activity, you’ll step into someone else’s shoes and experience how people with vision or hearing impairments might face challenges during an emergency.

Your Challenge:

Try to move from one side of the classroom to the other while one of your senses is blocked:

- Use a blindfold to block your sight

OR

- Use headphones or earplugs to block your hearing

Your teacher will give a warning signal using either a sound or a visual cue—but you won’t know which one ahead of time. Your job is to react as quickly and safely as possible.

Figure 4: Sensory Challenge Simulation in Emergency Situations.

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STEAM SCOUTS

DIY Lab

Objectives:

- Apply basic mechanical principles to design a device that communicates a warning in an alternative way.
- Build empathy by designing for users with different abilities.
- Test and refine a prototype through hands-on experimentation.

Possible Answers:

1. **What made it hard to detect the warning or know where to go?**
   - Limited visibility or unclear signals can make it difficult to identify where the warning is coming from. Noise or distractions might also hinder hearing the warning properly
2. **Which clues did you miss, and why?**
   - I might have missed visual or auditory clues because I was focused on other things or was too far away to notice the signs clearly.
3. **How would this feel in a real emergency?**
   - It would be very stressful and confusing to realize that I missed important warning signs. There could be panic as I try to figure out what to do next, especially if there's a time crunch.
4. **What could be added to help people with different needs respond faster?**
   - Different signals like flashing lights for those who are hearing impaired, clear visual signs, and loud, distinct alarms can be added to ensure everyone is aware of the emergency. Additionally, having clear, easy-to-follow maps and instructions can help all individuals act quickly

Activity Instructions:

1. **Set the Challenge** – Tell students they must design a simple machine (lever, pulley, wheel & axle, inclined plane, etc.) that can signal a warning without relying on light or sound.
2. **Form Groups** – Students work in pairs or small teams.
3. **Plan** – Use Think-Pair-Share to brainstorm different machine ideas.
4. **Gather Materials** – Provide classroom-safe building materials (cardboard, paper clips, string, small wheels, craft sticks, rubber bands, etc.).
5. **Build** – Students create their simple machine. Encourage quick prototyping and testing.
6. **Test in Challenge Mode** – One student acts as the user with limited vision or hearing (blindfold or earplugs), and another tries the evacuation task with their machine.
7. **Refine** – Based on feedback, groups make quick adjustments to improve function or safety.
8. **Share & Reflect** – Groups present their machine, explain how it works, and reflect on how it would help in a real emergency.

Possible Strategies for Implementation:

Strategy 1 – Think-Pair-Share

- How to Implement:

  At the start of the lab, have each student think silently for 1 minute about possible simple machine ideas that could signal a warning without light or sound. Then, pair up to discuss their ideas for 2 minutes. Finally, share the best ideas with the whole group so everyone benefits from multiple perspectives.

Strategy 2 – See-Think-Wonder (Visible Thinking Routine)

- How to Implement:

  After groups test their first prototype, display it for others to observe. Ask students to silently note:

  - See: What do you notice about the machine?
  - Think: What do you think it does and how does it help?
  - Wonder: What questions do you have or what could be improved?

  Groups then use this feedback to make quick refinements.

Strategy 3 – Round Robin

- How to Implement:

  During the reflection phase, have each student in the group share one improvement or unique feature of their design. Rotate speaking turns to ensure every voice is heard and valued.

Strategy 4 – Peer Testing and Feedback

- How to Implement:

  Groups exchange machines and act as “users” with limited vision or hearing (using blindfolds or earplugs). They test how easy it is to detect the warning signal. Afterwards, they give constructive feedback to the original designers.

Explain

Teacher’s Role:

- **Questioner**

  Prompt curiosity and guide thinking through open-ended questions:

- “If you couldn’t see or hear, what would you need to stay safe?”
- “What kinds of signals could a machine use that everyone could notice?”
- “How can we make sure our design works for people with different abilities?”

  Encourage deeper thinking by asking students to justify their design choices and predict possible problems.

- **Supporter**

  Model how to brainstorm and sketch design ideas, showing that rough ideas are a normal part of engineering. Offer visual prompts and sentence starters (e.g., "My design will work because..." or "Our machine will signal danger by...") to help students articulate their plans. Move around the room to check in with groups, troubleshoot technical issues, and ensure all students can participate meaningfully.

- **Collaborator**

  Actively join groups during the building and testing stages, asking guiding questions rather than giving direct solutions. Celebrate small successes and encourage peer-to-peer assistance. Reinforce teamwork by acknowledging when groups share materials, ideas, or problem-solving strategies.

- **Assessor**

  Observe students' planning, building, and testing processes. Take anecdotal notes on how they apply mechanical concepts and whether they're considering the needs of different users. Use quick check-in questions to assess understanding (e.g., "What makes your signal noticeable?" or "Who could benefit from this design?"). Provide feedback that encourages iteration and refinement.

Expected From Students:

- Watch the video attentively and note examples of inclusive emergency systems.
- Connect the video examples to their own DIY lab designs.
- Suggest at least one improvement to their prototype based on insights from the video.
- Actively participate in group discussion and reflection.

Parts of the book included.

Screen Lessons

Watch this video to explore how emergency planning can support people with disabilities:

https://youtu.be/phzNF4wwLs0?si=V0wPg77UuKGxxP6d

Screen Lessons

Objective:

Understand how inclusive emergency planning supports people with disabilities and apply this understanding to refine a prototype.

Video link: https://youtu.be/phzNF4wwLs0?si=5dNHS4-bfccmorTd

Possible Strategies for implementation:

1. **Pre-Watching: Set a Purpose**

   - Strategy: Think-Pair-Share

How to Implement:

- Ask students to think individually about what features they expect to see in an inclusive emergency system.
- Pair with a partner to share ideas.
- Share with the class to create a "Look For" checklist on the board before watching.

2. During Watching: Active Engagement

- Strategy: See-Think-Wonder (Visible Thinking Routine)

How to Implement:

1. Students jot down:
   - See: What examples or devices are shown?
   - Think: Why might these be effective for different needs?
   - Wonder: What else could be improved or added?
2. Pause the video briefly at key points for quick note-taking.

3. Post-Watching: Discussion and Reflection

- Strategy: Round Robin

How to Implement:

- In small groups, each student shares one idea from the video that could improve their prototype.
- Rotate until everyone has shared.
- Groups select their top improvement idea to present to the class.

Dig Deeper

Objective:

Understand the potential dangers of gas stoves and appliances, and connect this information to safety measures discussed in class.

Article link: https://www.propublica.org/article/what-to-know-about-gas-stove-risks?utm_source=chatgpt.com

KWL Chart Link: https://g.co/gemini/share/9a5ceefdf570

1. Pre-Reading: Set a Purpose

- Strategy: Think-Pair-Share

How to Implement:

1. Ask students to think silently for 1 minute about what dangers they already know (or think they know) about gas stoves.
2. Pair up to share ideas and predictions.
3. Share key points as a class to create a "What We Expect to Learn" list before reading.

2. **During Reading: Active Engagement**
   - Strategy:

How to Implement:

1. Students highlight key facts in one color and potential safety solutions in another color.
2. Use margin notes or sticky notes to jot quick questions or surprising facts.
3. Pause halfway through to let students share one fact and one question with a partner.

3. **Post-Reading: Discussion and Reflection**
   - Strategy: Compass Points (Visible Thinking Routine)

How to Implement:

In small groups, students respond to four prompts:

- **E (Excited):** What new information excited you?
- **W (Worrisome):** What worries you about what you read?
- **N (Need to Know):** What more do you want to know?
- **S (Suggestions):** What safety actions do you suggest?

Groups present one key idea from each category to the class.

Elaborate and Evaluate

Teacher's Role:

- **Supporter**

  Guide students in brainstorming and sketching design ideas. Provide sentence starters and prompts such as "Our machine will work because..." or "The system will help by..." Walk around the room to offer support to students and facilitate discussions as needed.

- **Collaborator**

  Actively participate in group work, offering guidance and encouraging peer collaboration. Reinforce teamwork by acknowledging when students share materials or help one another troubleshoot their designs.

- **Assessor**

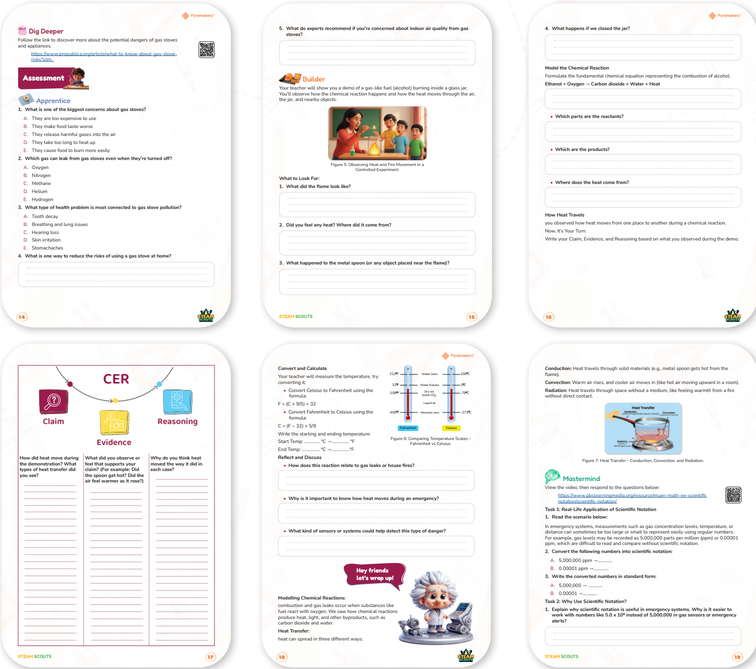
  Observe students as they work through the building and testing stages. Use informal assessment strategies to gauge their understanding of the concepts and to see if they're applying critical thinking. Use probing questions to assess their design choices and the improvements they make.

Expected from students:

- Collaborate in groups to design and build a machine that detects danger without relying on sound or light.

- Test their prototypes and offer feedback to peers for improvement.
- Reflect on how their designs could help people in real-world situations like gas leaks or fire emergencies.

Part of the book Included.



Apprentice

Objective:

To assess student comprehension and knowledge of the concepts introduced in the DIY lab and video.

Possible answers:

1. **What is one of the biggest concerns about gas stoves?**
   - Answer: C. They release harmful gases into the air
2. **Which gas can leak from gas stoves even when they're turned off?**
   - Answer: C. Methane
3. **What type of health problem is most connected to gas stove pollution?**
   - Answer: B. Breathing and lung issues
4. **What is one way to reduce the risks of using a gas stove at home?**
   - One way to reduce the risks is to ensure proper ventilation by turning on an exhaust fan or opening windows while cooking. This helps remove harmful gases such as carbon monoxide and nitrogen dioxide from the air.
5. **What do experts recommend if you're concerned about indoor air quality from gas stoves?**
   - Experts recommend using a range hood that vents outdoors, regularly maintaining

and checking your stove for leaks, and considering switching to an electric or induction cooktop to reduce indoor air pollution.

Possible Strategies for implementation:

1. **Pre-Assessment Review**
   - Review key facts about gas stove safety from the lesson before students take the quiz.
   - Use a quick "Think-Pair-Share" where students recall potential dangers and solutions before answering.
2. **Independent Assessment**
   - Have students complete the questions individually to check their understanding.
   - Remind them to read each question carefully and consider all options before choosing.
3. **Peer Discussion & Correction**
   - After answering, pair students to compare and discuss their answers.
   - Encourage them to explain why they chose each answer to reinforce reasoning skills.
4. **Whole-Class Feedback**
   - Go through each question as a class, asking volunteers to share their answers and reasoning.
   - Provide real-life examples to clarify misconceptions (e.g., how methane leaks can happen).
5. **Connection to Real Life**
   - Ask students to relate each correct answer to real-world safety actions they can take at home.
   - For example: "If methane can leak from a stove, what should you do if you smell gas?"
6. **Extension Activity**
   - Have students design a simple safety poster for kitchens that includes tips learned from the quiz.
   - Posters can be shared around the school or at home.

Builder

Objective:

Students will evaluate and enhance their DIY projects by observing a demonstration, identifying key concepts, and analyzing how heat transfers in a chemical reaction.

possible answers:

What to Look For

1. **What did the flame look like?**
   - Bright yellow-orange flame, flickering inside the jar.
   - Possibly blue edges if burning clean alcohol.

2. **Did you feel any heat? Where did it come from?**
    - Yes, heat came from the burning fuel.
    - Heat radiated outward through the jar and into the surrounding air.
3. **What happened to the metal spoon (or any object placed near the flame)?**
    - It got warmer or hot after a few seconds due to heat transfer (mainly conduction).
4. **What happens if we closed the jar?**
    - When the jar was closed, the flame slowly went out because it used up the oxygen inside and could no longer keep burning.

Model the Chemical Reaction

- **Reactants:** Ethanol (C_2H_5OH) + Oxygen (O_2)
- **Products:** Carbon dioxide (CO_2) + Water (H_2O) + Heat

Where does the heat come from?

- From the energy released when chemical bonds in ethanol and oxygen break and new bonds form in carbon dioxide and water.

CER Section – How Heat Travels

- **Claim:** Heat from the flame moved through the air, the jar, and to objects nearby by conduction, convection, and radiation.
- **Evidence:** The spoon became hot (conduction), warm air rose above the flame (convection), and you could feel heat from a distance (radiation).
- **Reasoning:** Heat transfer occurs in three main ways, and in a burning flame, all are present—energy is transferred through materials, moving air, and electromagnetic waves.

Temperature Conversion Example

If start temp was 25°C:

- 25°C → (25 × 9/5) + 32 = 77°F

If end temp was 40°C:

- 40°C → (40 × 9/5) + 32 = 104°F

Reflect and Discuss

1. **How does this reaction relate to gas leaks or house fires?**
    - Gas leaks can ignite and burn similarly, producing heat and dangerous gases.
    - Understanding combustion helps in fire prevention and safety.
2. **Why is it important to know how heat moves during an emergency?**
    - It helps predict fire spread, escape safely, and position alarms or detectors effectively.
3. **What kind of sensors or systems could help detect this type of danger?**
    - Smoke detectors, gas leak detectors (methane sensors), heat sensors, flame detectors, and fire alarms.

Materials Needed:

Alcohol-based fuel

- Glass jar
- Metal spoon
- Lighter or matches
- Thermometer

Activity Instructions:

- **Introduce the Concept**

  Show the students a demonstration of a gas-like fuel burning inside a jar. Explain the chemical reaction and how heat moves through materials and air.

- **Set the Purpose**

  Explain to students that they will observe how heat moves during a combustion reaction and how it relates to gas stove safety and heat detection systems.

- **Guide Access to the Game**

  Guide students through a about heat transfer, asking them to classify different heat transfer methods (conduction, convection, and radiation).

- **Support Student Engagement**

  Help students answer questions during the experiment:

  - What did the flame look like?
  - Where did you feel the heat?
  - How did the heat affect objects nearby?
  - What happens during combustion?

Mastermind

Objective:

Students will apply the concepts of scientific notation and its relevance in emergency systems, particularly in measuring gas concentrations, and learn how scientific notation helps with better interpretation and comparison of large and small numbers in real-world contexts.

Possible Answers:

Task 1: Real-Life Application of Scientific Notation

1. **Convert the following numbers into scientific notation:**

   A. 5,000,000 ppm → 5.0×10^6 ppm

   B. 0.00001 ppm → 1.0×10^{-5} ppm

2. **Write the converted numbers in standard form:**

   A. 5.0×10^6 → 5,000,000

B. $1.0 \times 10^{-5} \rightarrow 0.00001$

Task 2: Why Use Scientific Notation?

1. Explain:

Scientific notation is useful in emergency systems because it allows very large or very small numbers to be written in a shorter, clearer form, making them easier to read, compare, and communicate quickly—important when responding to gas sensor alerts or emergencies.

2. Compare two measurements:

- Gas level A: 3.4×10^4 ppm = 34,000 ppm
- Gas level B: 0.00012×10^4 ppm = 1.2 ppm

Answer: Gas level A is much higher. Scientific notation helps by clearly showing the order of magnitude and making the size difference obvious.

Task 3: Use in Emergency Systems

- 5.2×10^4 ppm → 52,000 ppm
- 3.0×10^5 ppm → 300,000 ppm

Explanation: Using scientific notation makes it easier to compare values, record them accurately, and communicate dangerous levels quickly—critical during emergencies.

Bonus Challenge

Starting level: 1.0×10^2 ppm (100 ppm)

- After 2 minutes → × 10 → 1.0×10^3 ppm (1,000 ppm)
- After 4 minutes → × 10 again → 1.0×10^4 ppm (10,000 ppm)

Final Answer: 1.0×10^4 ppm

Activity Instructions:

In-Class Introduction

- **Activate Prior Knowledge:**
  - Ask: "Where have you seen very large or very small numbers in real life?"
  - Encourage students to think of examples like space distances, bacteria size, or pollution levels.
- **Introduce the Context:**
  - Explain that in emergency systems, such as gas leak monitoring, numbers can be extremely large or small. Scientific notation helps record and compare them quickly.
- **Show the Video:**
  - Play the PBS video. Ask students to focus on why scientific notation is useful.

At-Home Component (Independent Work)

- Students will complete the worksheet tasks:
  1. Convert large and small numbers between standard form and scientific notation.
  2. Compare different gas concentration levels to determine which is higher.

3. Explain why scientific notation is important in emergencies.
4. Apply to a real-world example with carbon monoxide levels.
5. Bonus Challenge: Predict the increase in a gas leak over time using multiplication by powers of ten.

Classroom Follow-Up (Next Lesson, 10 minutes)

- Review Answers: Discuss the correct conversions and reasoning from the answer key.
- Highlight Misconceptions: Clarify common errors (e.g., incorrect placement of decimal point, misunderstanding negative exponents).
- Connect to Safety: Emphasize how these skills help engineers, scientists, and emergency responders act quickly and accurately.

Differentiation Tips

- For Support: Allow students to use a step-by-step conversion guide.
- For Challenge: Ask students to create their own emergency data set in both forms for peers to solve.

Now I Can

Objective:

Students will reflect on their learning journey throughout the lesson, identifying and articulating their mastery of key concepts related to the lesson, such as the application of simple machines or scientific notation. They will engage in self-assessment, evaluate their understanding of core concepts, and collaborate with peers to deepen their comprehension.

Activity Instructions:

Step 1: Revisit the Lesson's Work

- **Objective:** To set the stage for reflection by revisiting key elements of the lesson.
- **Instructions:**

1. Ask students to review their work from the lesson (e.g., diagrams, designs, calculations, or notes).
2. Provide guiding questions such as:
   - o "What were the main concepts you learned today?"
   - o "How did the activity help you understand the concepts more clearly?"
   - o "What was the most challenging part of the lesson, and why?"
3. Give students 10–5 minutes to reflect silently before moving to the next step. This allows them time to consolidate their thinking.

Step 2: Student Self-Reflection

- Objective: To encourage students to evaluate their own understanding of the lesson and identify areas of strength and improvement.

- Instructions:
  1. Have each student complete a short self-reflection prompt on their own, such as:
     - “What did I learn today?”
     - “How well do I feel I understood the concept of [key lesson concept]?”
     - “What would I do differently if I had to repeat the activity?”
  2. Optionally, use a scale (e.g., 5-1 or 10-1) where students rate their confidence in specific learning outcomes, such as “Understanding the application of simple machines” or “Using scientific notation in real-life examples.”
  3. Allow students 15–10 minutes for this reflection. It can be done individually or digitally for easy sharing.

Step 3: Small Group Discussion

- Objective: To promote peer-to-peer learning and provide students with a platform to discuss their reflections and deepen their understanding through discussion.
- Instructions:
  1. After students have completed their self-reflection, have them form small groups (4–3 students per group).
  2. In their groups, students will share their reflections on the lesson. Encourage them to discuss:
     - “What was one key thing you learned today that you didn’t know before?”
     - “How did your understanding change as a result of today’s activities?”
     - “What strategies worked best for you when completing the tasks?”
  3. After discussing in small groups, have a brief class-wide discussion where each group shares one key takeaway from their conversation. This will allow students to learn from each other’s perspectives and further solidify their understanding.

Relations to standard:

1. ISTE Standards for Students

ISTE Standard 1: Empowered Learner

- Students take an active role in their learning by setting goals, reflecting on progress, and making adjustments to improve.

  Application: Step 2 (Self-Reflection) encourages students to reflect on their learning, set goals, and identify areas of growth.

ISTE Standard 6: Creative Communicator

- Students communicate clearly and creatively using a variety of digital tools and formats.

  Application: During Step 3 (Small Group Discussion), students communicate their ideas and insights, collaborating creatively with peers.

2. NGSS (Next Generation Science Standards)

NGSS Science and Engineering Practice 1: Asking Questions and Defining Problems

- Students ask questions to investigate problems and explore solutions based on their observations.

  Application: The self-reflection process (Step 2) prompts students to think critically about their learning and identify questions related to the lesson's core concepts.

NGSS Science and Engineering Practice 7: Engaging in Argument from Evidence

- Students construct and defend arguments based on evidence collected through investigations and experimentation.

  Application: Step 3 (Small Group Discussion) encourages students to engage in discussions, share evidence, and build their arguments based on what they learned during the lesson.

3. Common Core State Standards (CCSS) for English Language Arts

CCSS.ELA-LITERACY.SL.8.1: Comprehension and Collaboration

- Students engage in collaborative discussions, building on others' ideas and expressing their own ideas clearly.

  Application: In Step 3 (Small Group Discussion), students engage with one another, offering insights and reflecting on their learning while fostering collaborative conversations.

CCSS.ELA-LITERACY.W.8.10: Write Routinely

- Students write routinely for a range of tasks, purposes, and audiences.

  Application: Step 2 (Self-Reflection) asks students to write about their learning, connecting their experiences to the lesson's concepts and reinforcing writing as a reflective tool.

4. CSTA (Computer Science Teachers Association) Standards

CSTA Standard 1A-AP-08: Design a Solution to a Problem

- Students design solutions to real-world problems, applying principles of engineering and computational thinking.

  Application: The DIY Lab activity allows students to design simple machines that address real-world problems, such as improving safety with gas stoves.

5. SEL (Social and Emotional Learning) Standards

SEL Competency 1: Self-Awareness

- Students recognize and understand their emotions, strengths, and areas for growth.

  Application: In Step 2 (Self-Reflection), students assess their understanding of the lesson, fostering self-awareness about their learning.

SEL Competency 4: Relationship Skills

- Students establish and maintain healthy relationships, communicating effectively, and resolving conflicts.

  Application: Step 3 (Small Group Discussion) provides students with the opportunity to practice effective communication and collaboration, supporting relationship-building skills.